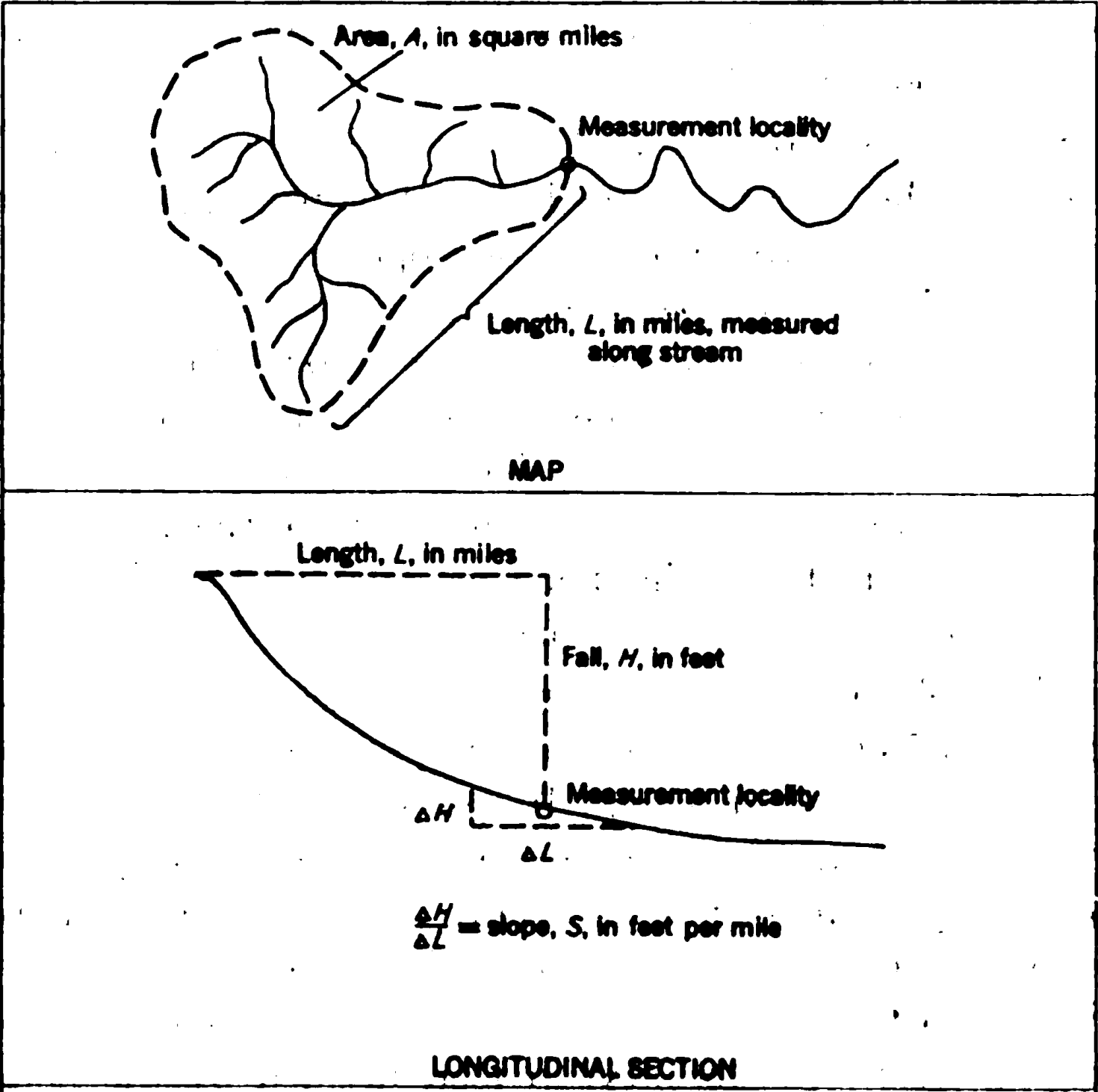


The Stream Power Law

Introduction

Jean Braun 2024

Hack, 1957



Studies of Longitudinal Stream Profiles in Virginia and Maryland

By JOHN T. HACK
SHORTER CONTRIBUTIONS TO GENERAL GEOLOGY
GEOLOGICAL SURVEY PROFESSIONAL PAPER 294-B

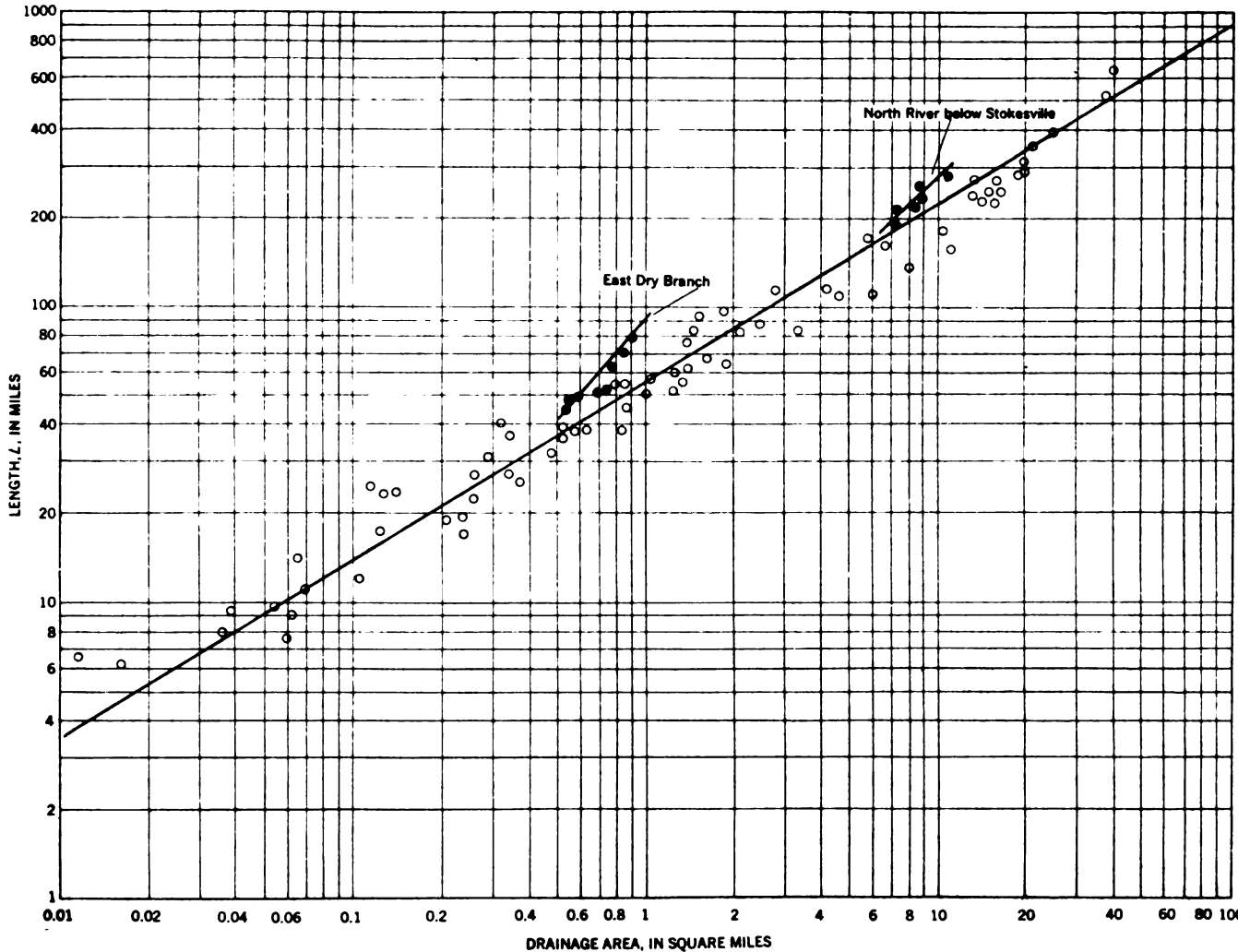


FIGURE 25.—Logarithmic scatter diagram showing the relation between length and drainage area at all the measurement localities. Localities in East Dry Branch below the sandstone area and on the North River are shown in solid black.

$$Q \propto A$$

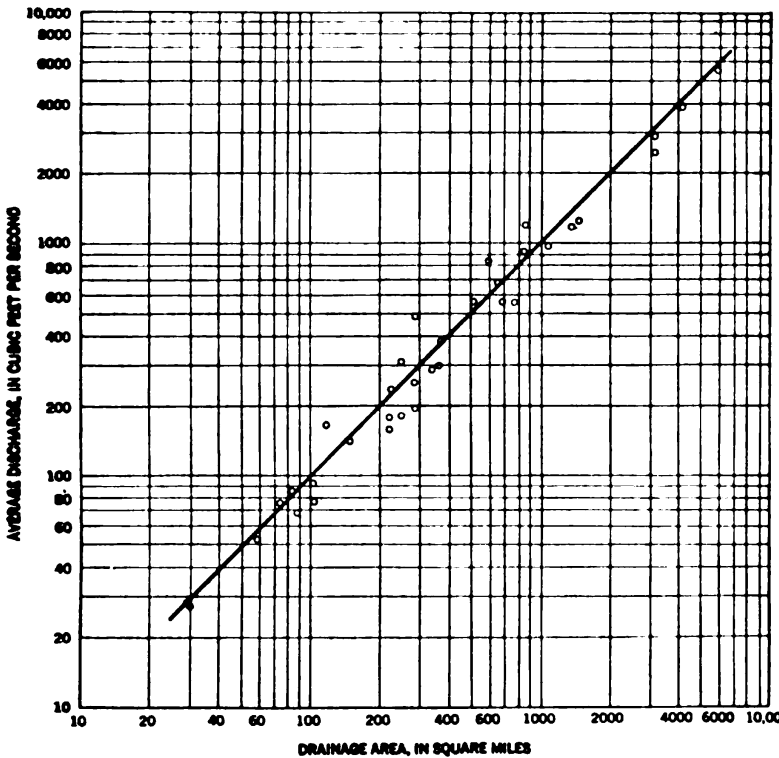


FIGURE 15.—Logarithmic graph showing the relations between drainage area and discharge at gaging stations in the Potomac River basin above Washington, D.C.

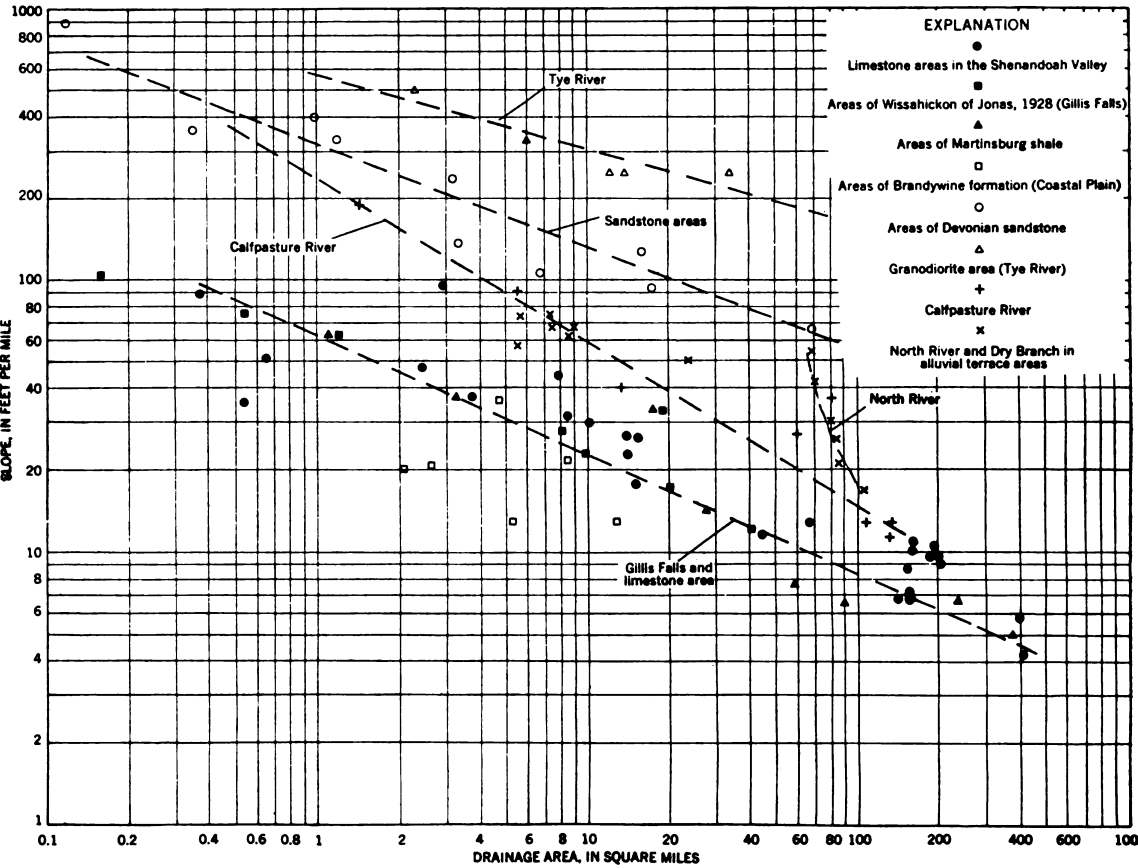


FIGURE 16.—Logarithmic scatter diagram showing the relation between channel slope and drainage area at the measurement localities (table 8).

$$\log L = \log k_r + r \log A$$

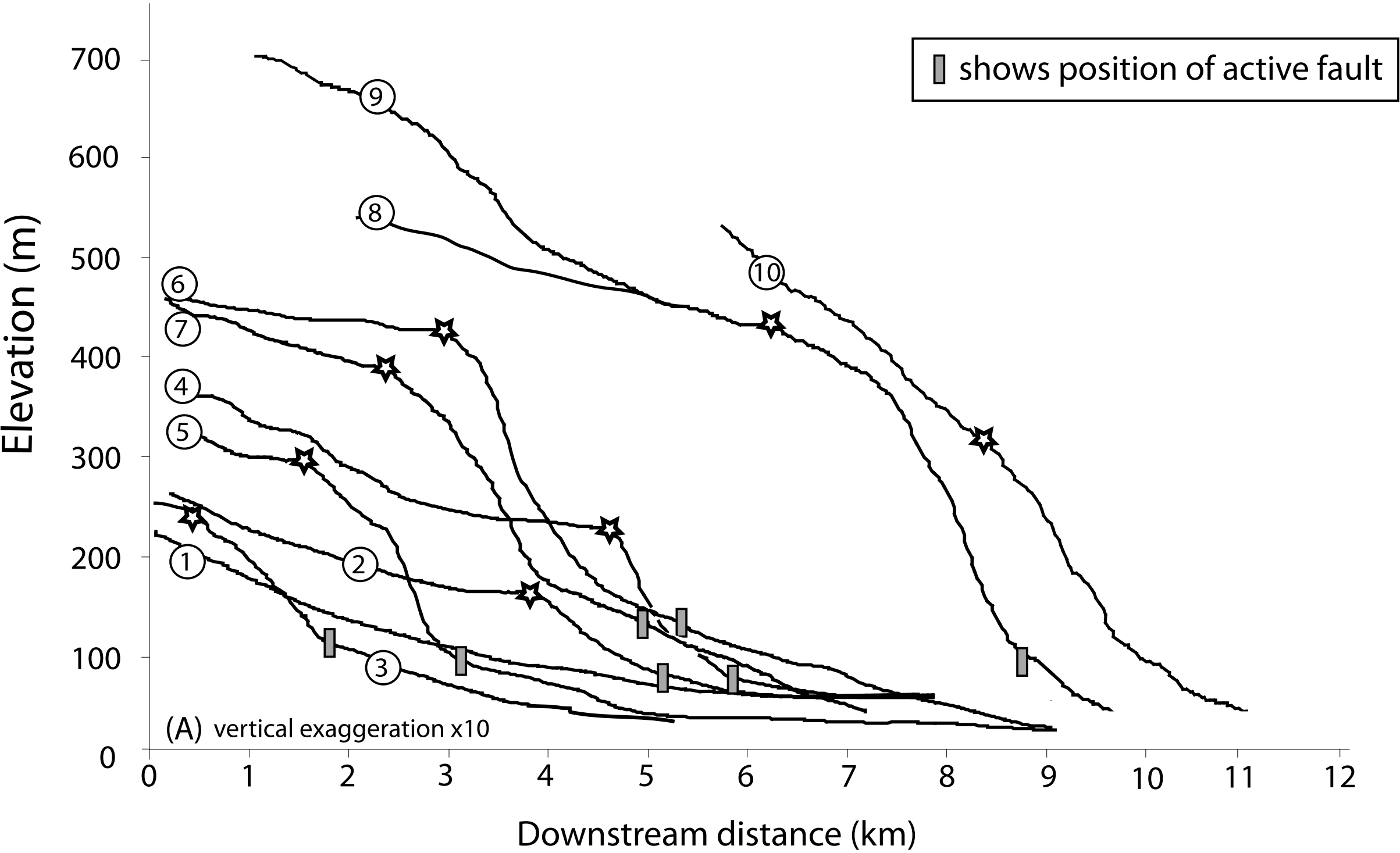
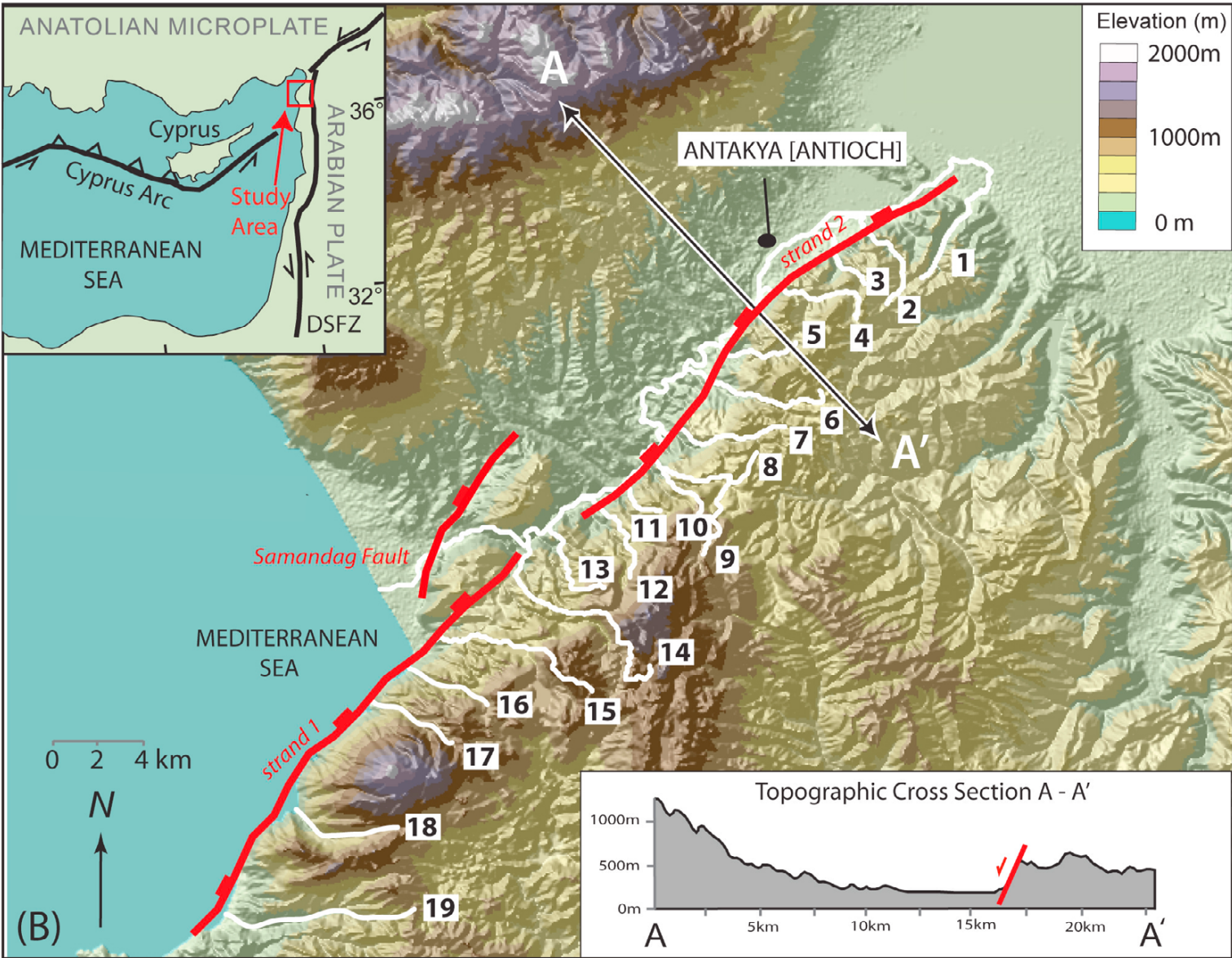
$$\log S = \log k_c - c \log A$$

Knickpoints

JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 117, F02024, doi:10.1029/2011JF002157, 2012

Tectonic and climatic controls on knickpoint retreat rates and landscape response times

Alexander C. Whittaker¹ and Sarah J. Boulton²



The Stream Power Law

$$\frac{\partial h}{\partial t} = U - K\phi^m S^n$$

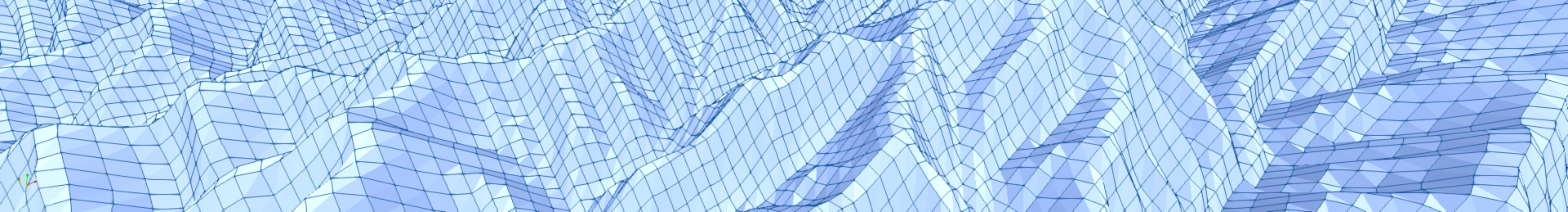
Height $\rightarrow \partial h$
 Time $\rightarrow \partial t$
 Uplift rate $\rightarrow U$
 Constant $\rightarrow K$
 Discharge $\rightarrow \phi$
 Exponents $\rightarrow m, n$
 Slope $\rightarrow S$

$$S = \frac{\partial h}{\partial x} \quad \phi = \int_A \nu \, dA = A\bar{\nu}$$

Precipitation rate $\rightarrow \nu$

$$\frac{\partial h}{\partial t} = U - K_f A^m \left(\frac{\partial h}{\partial x} \right)^n$$

Whipple and Tucker, 1999



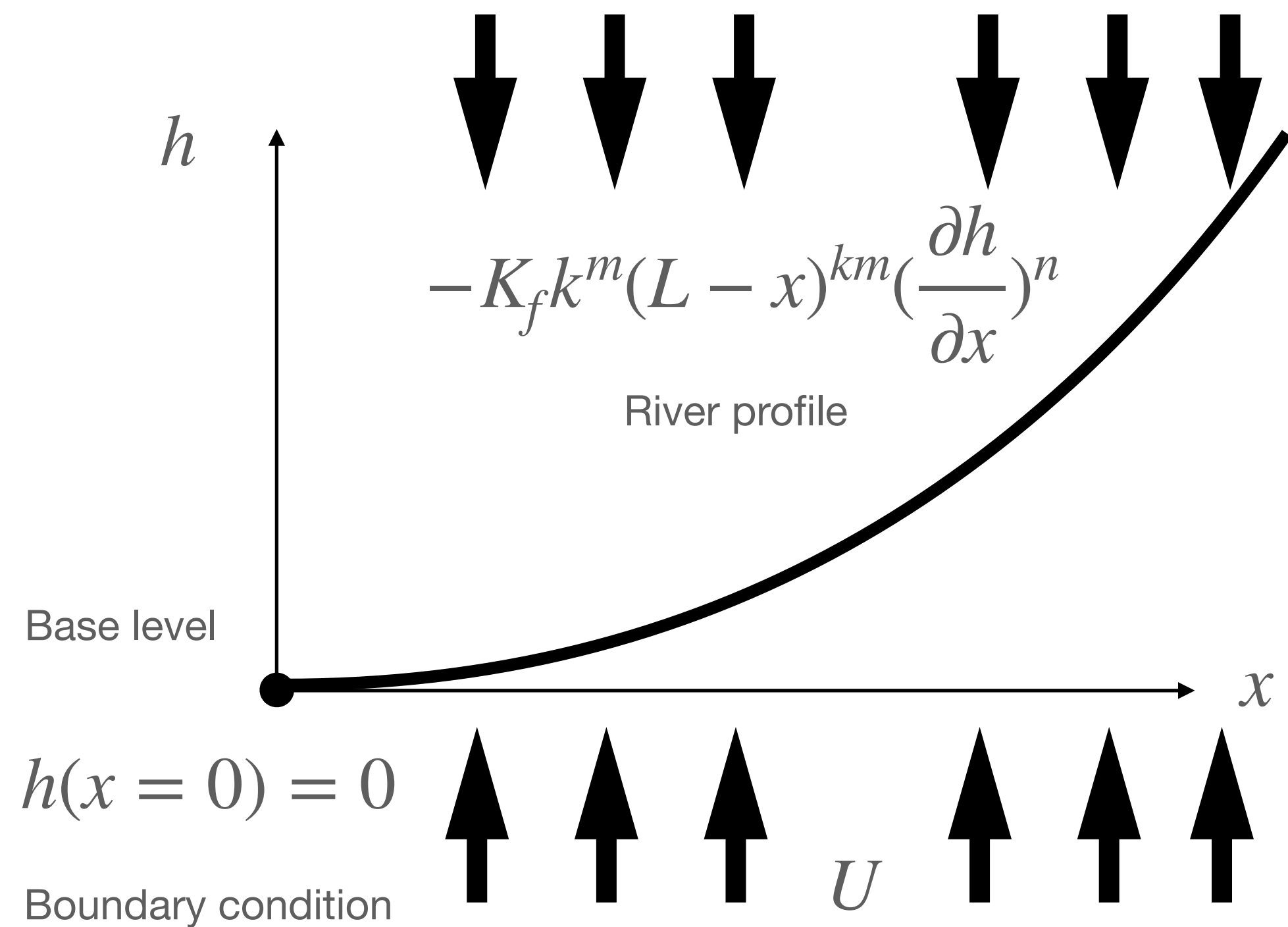
Drainage area: Hack's law

$$A = k(L - x)^p$$

$$\frac{\partial h}{\partial t} = U - K_f k^m (L - x)^{km} \left(\frac{\partial h}{\partial x} \right)^n$$

Hack, 1957

Steady-state solution - river profiles



Whipple and Tucker, 1999

Steady-state

$$\frac{\partial h}{\partial t} = U - K_f k^m (L - x)^{mp} \left(\frac{\partial h}{\partial x}\right)^n = 0$$

$$\frac{\partial h}{\partial x} = \left(\frac{U}{K_f k^m}\right)^{1/n} (L - x)^{-mp/n}$$

$$h = C - \left(\frac{U}{K_f k^m (1 - mp/n)}\right)^{1/n} (L - x)^{1 - mp/n}$$

$$h = \left(\frac{U}{K_f k^m (1 - mp/n)}\right)^{1/n} [L^{1 - mp/n} - (L - x)^{1 - mp/n}]$$

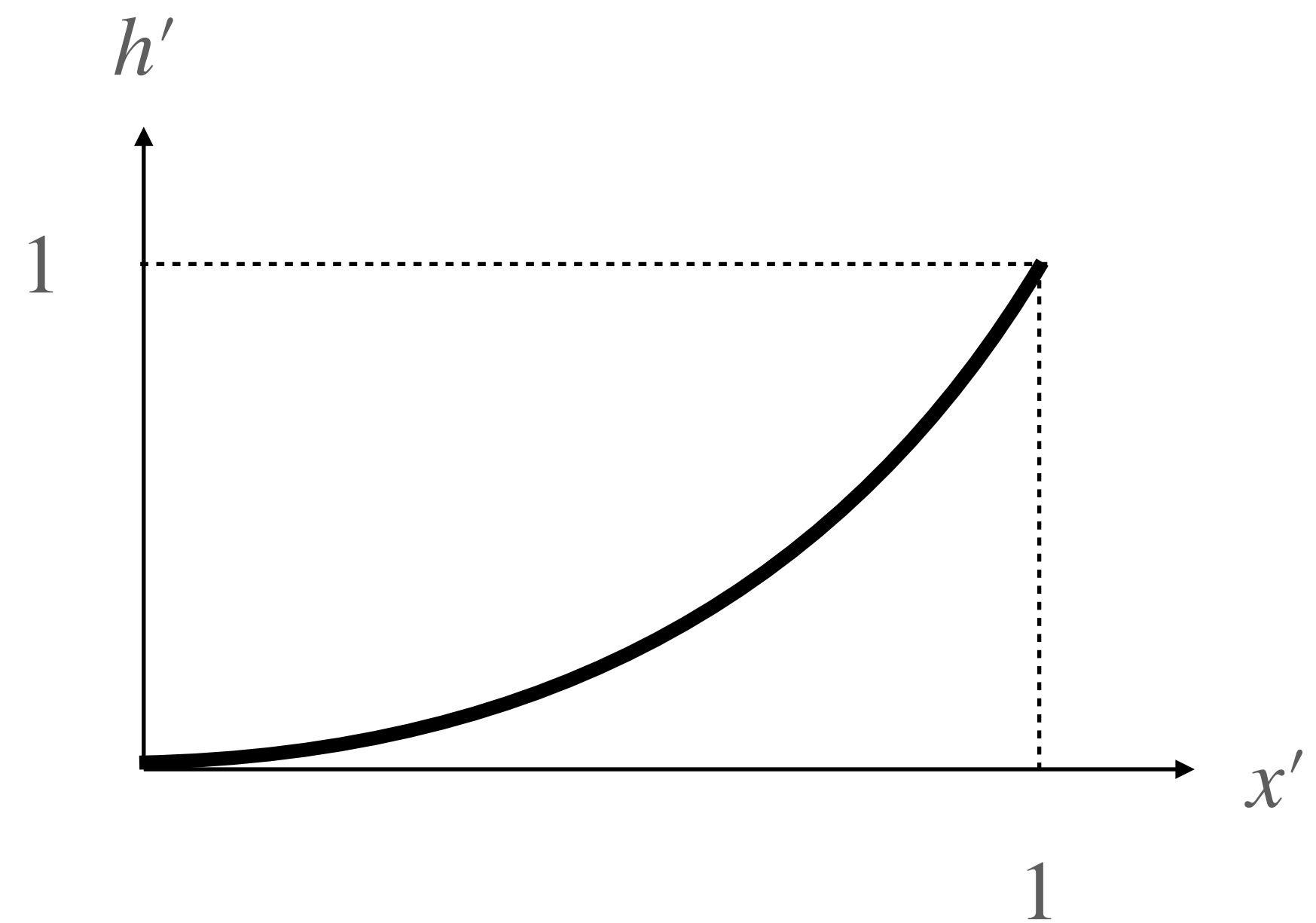
Adimensional solution

$$h = \left(\frac{U}{K_f k^m (1 - mp/n)} \right)^{1/n} [L^{1-mp/n} - (L - x)^{1-mp/n}]$$

$$h'(x') = 1 - (1 - x')^{1-mp/n}$$

$$x' = x/L \quad h' = h/h_{\max}$$

$$h_{\max} = \left(\frac{U}{K_f k^m (1 - mp/n)} \right)^{1/n} L^{1-mp/n}$$



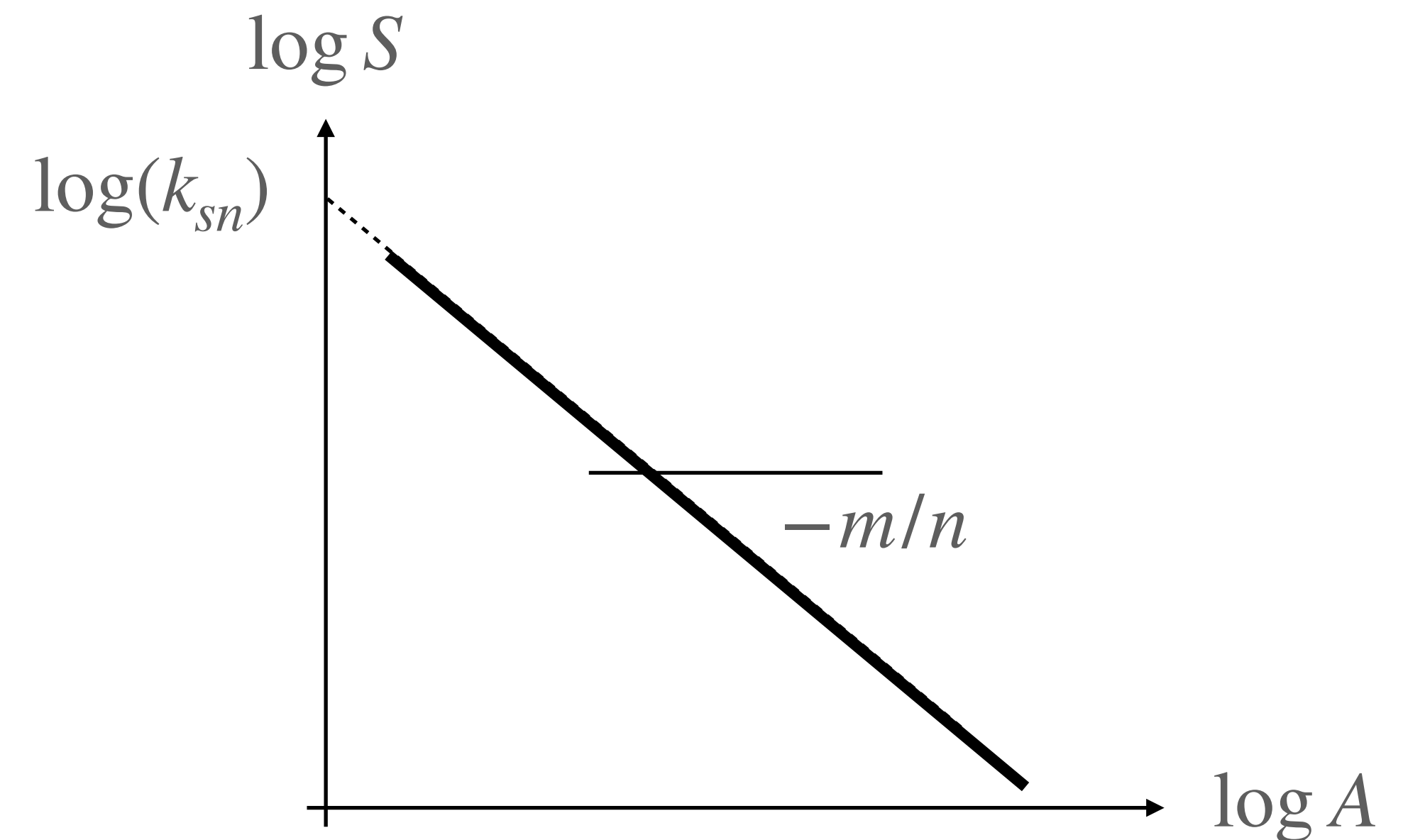
Whipple and Tucker, 1999

Slope-area relationship

$$\frac{\partial h}{\partial t} = U - K_f A^m S^n = 0$$

$$S = \left(\frac{U}{K_f}\right)^{1/n} A^{-m/n}$$

$$\log S = \log(k_{sn}) - \theta \log A$$



Flint, 1974

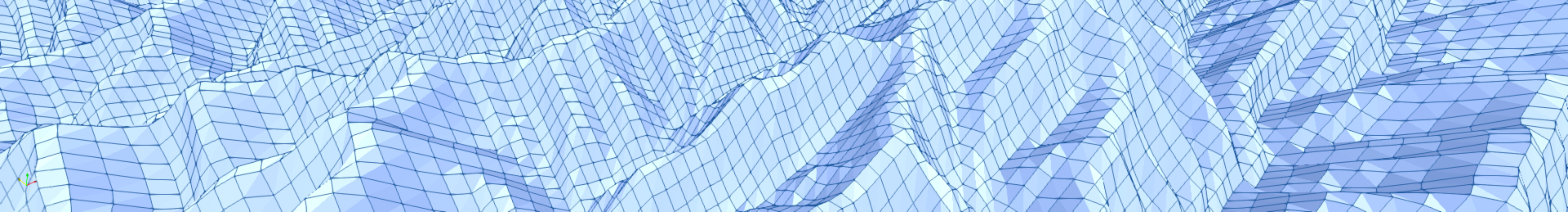
Concavity

$$\theta = \frac{m}{n}$$

$$k_{sn} = \left(\frac{U}{K_f}\right)^{1/n}$$

Channel steepness

$$k_{sn} = SA^\theta$$



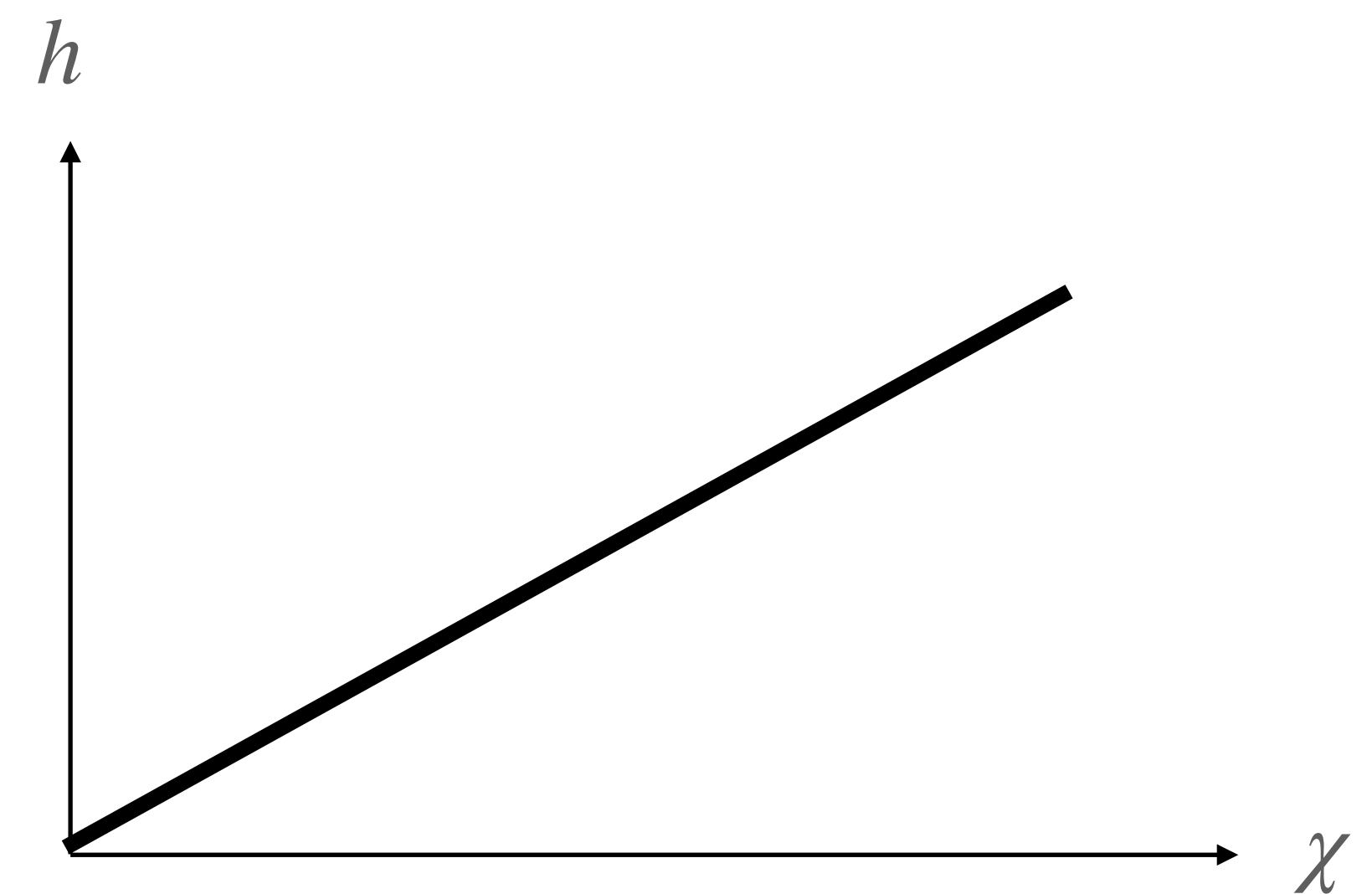
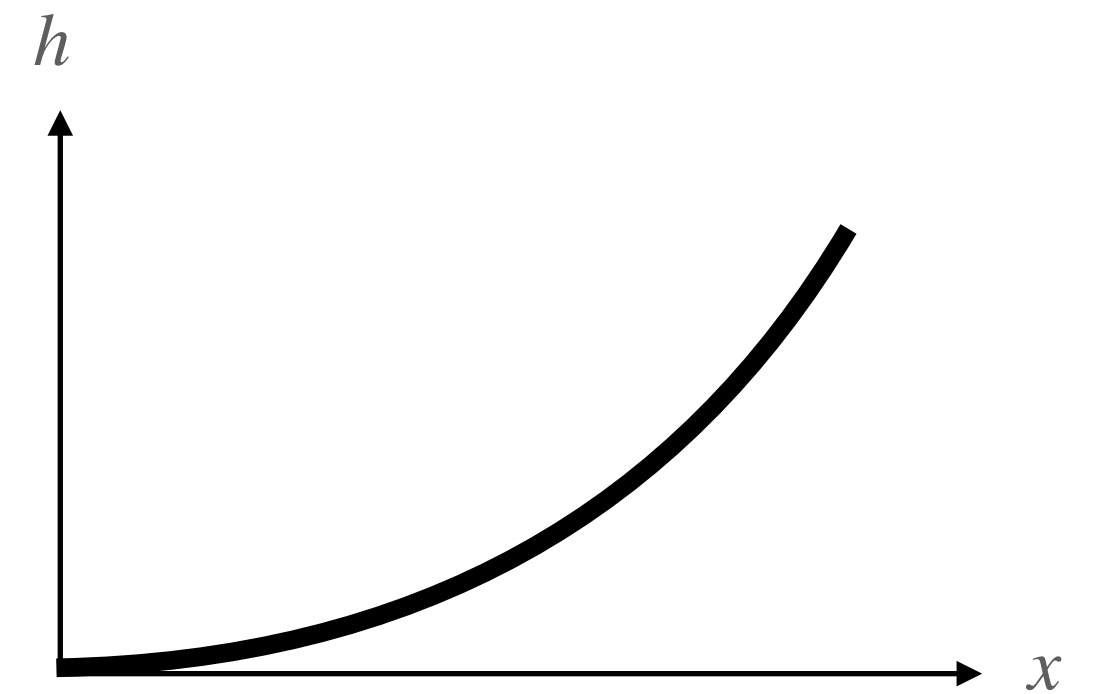
χ (chi)

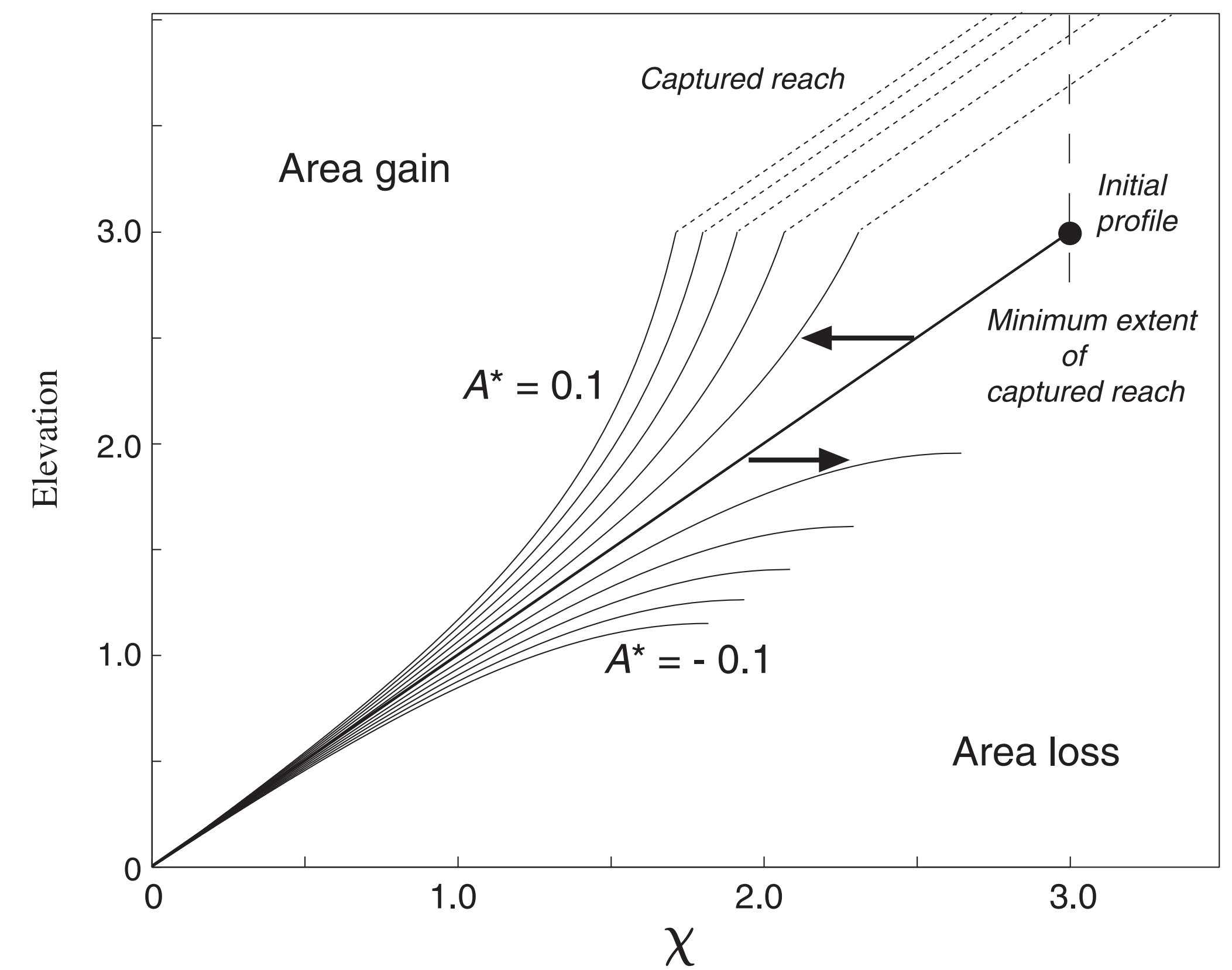
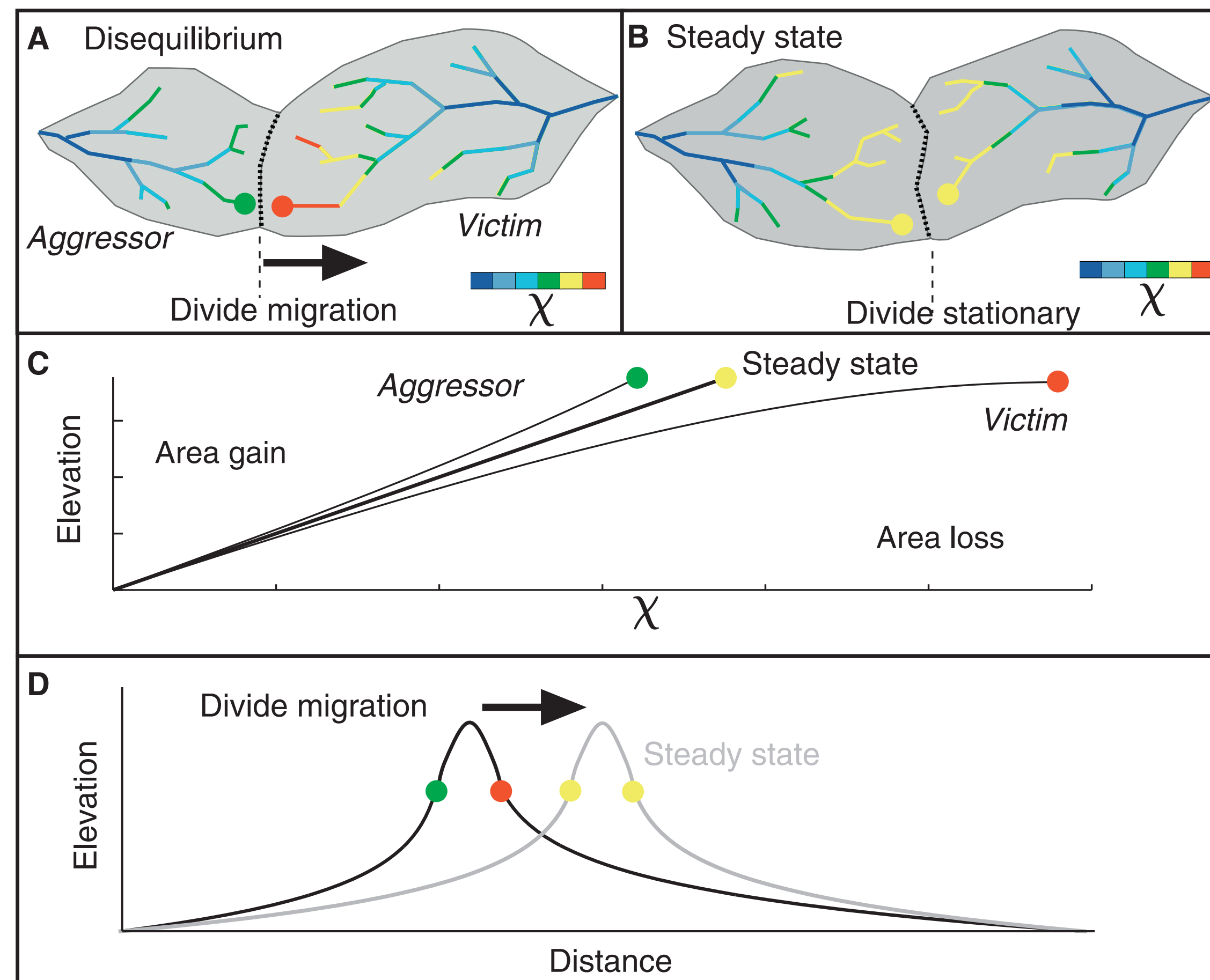
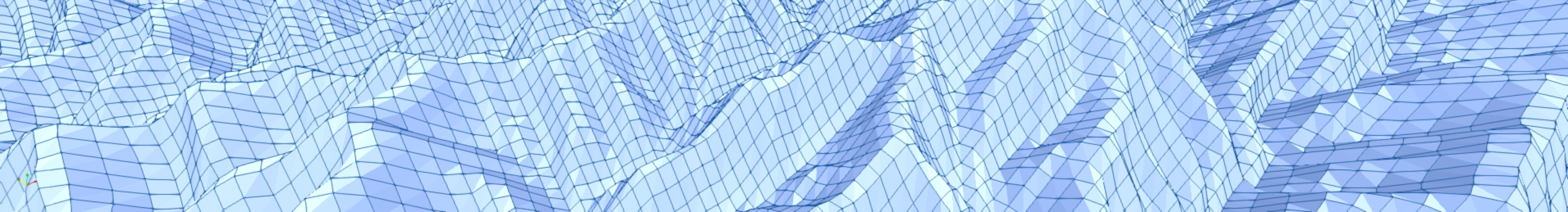
$$\frac{\partial h}{\partial t} = U - K_f A^m \left(\frac{\partial h}{\partial x} \right)^n = 0$$

$$h(x) = \left(\frac{U}{K_f A_0} \right)^{1/n} \chi$$

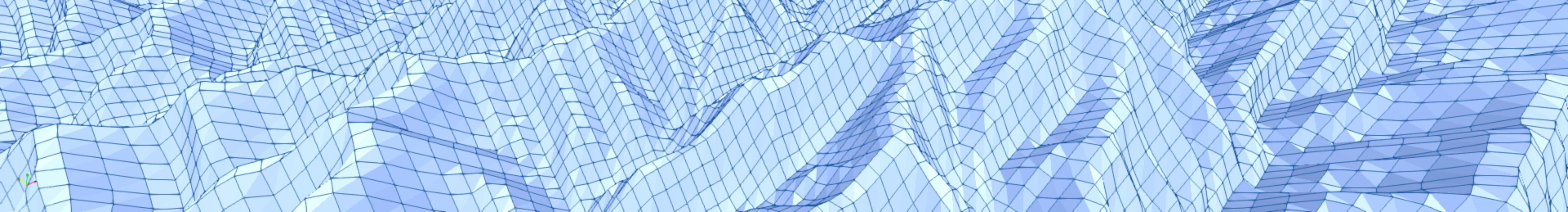
$$\chi = \int_0^x \left(\frac{A_0}{A(x')} \right)^{m/n} dx'$$

Royden and Perron, 2013





Willett et al, 2014



Finite Difference Solution

$$\frac{\partial f}{\partial x} = \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon) - f(x)}{\epsilon}$$

Finite Difference approximation:

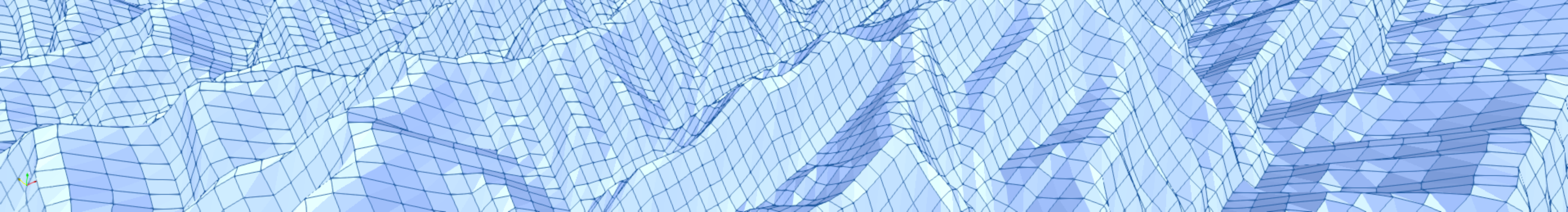
$$\frac{\partial f}{\partial x} \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Taylor series' expansion

$$f(x + \Delta x) = \sum_{n=0}^{\infty} \frac{\Delta x^n}{n!} \frac{\partial^n f}{\partial x^n} = f(x) + \Delta x \frac{\partial f}{\partial x}(x) + \epsilon(\Delta x^2)$$

Order of the approximation

$$\frac{\partial f}{\partial x} = \frac{f(x + \Delta x) - f(x)}{\Delta x} + \epsilon(\Delta x)$$



Upwind scheme

Forward-difference scheme

$$\frac{\partial f}{\partial x} = \frac{f(x + \Delta x) - f(x)}{\Delta x} + \epsilon(\Delta x)$$

There exists another scheme

$$f(x - \Delta x) = \sum_{n=0}^{\infty} \frac{(-\Delta x)^n}{n!} \frac{\partial^n f}{\partial x^n} = f(x) - \Delta x \frac{\partial f}{\partial x}(x) + \epsilon(\Delta x^2)$$

Backward-difference scheme

$$\frac{\partial f}{\partial x} = \frac{f(x) - f(x - \Delta x)}{\Delta x} + \epsilon(\Delta x)$$

It has the same level of approximation than the forward-difference scheme

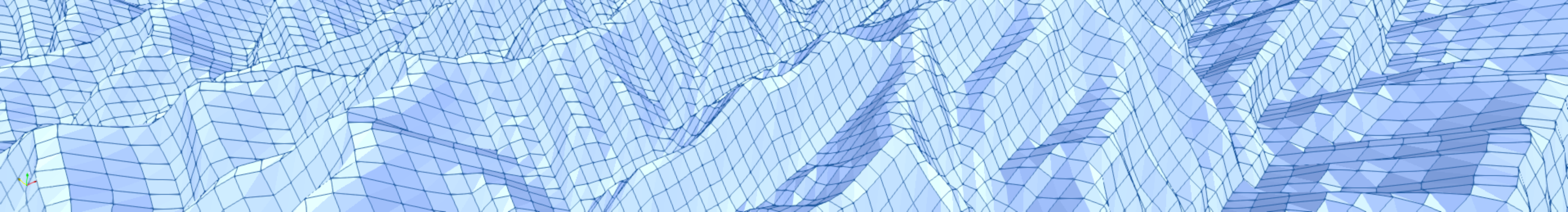
Second-order approximation

$$(1) \quad f(x + \Delta x) = \sum_{n=0}^{\infty} \frac{\Delta x^n}{n!} \frac{\partial^n f}{\partial x^n} = f(x) + \Delta x \frac{\partial f}{\partial x}(x) + \frac{\Delta x^2}{2} \frac{\partial^2 f}{\partial x^2}(x) + \epsilon(\Delta x^3)$$

$$(2) \quad f(x - \Delta x) = \sum_{n=0}^{\infty} \frac{(-\Delta x)^n}{n!} \frac{\partial^n f}{\partial x^n} = f(x) - \Delta x \frac{\partial f}{\partial x}(x) + \frac{\Delta x^2}{2} \frac{\partial^2 f}{\partial x^2}(x) + \epsilon(\Delta x^3)$$

$$(1-2) \quad \frac{\partial f}{\partial x} = \frac{f(x + \Delta x) - f(x - \Delta x)}{2\Delta x} + \epsilon(\Delta x^2)$$

Centered difference approximation



Upwind scheme

The SPL is an equation of the general form of the advection equation

$$\frac{\partial h}{\partial t} = -K_v A^m \frac{\partial h^n}{\partial x} \quad \longleftrightarrow \quad \frac{\partial f}{\partial t} = v \frac{\partial f}{\partial x}$$

If v is positive we must use a forward scheme, if v is negative we must use a backward difference

$$\frac{\partial f}{\partial x} = \frac{f(x + \Delta x) - f(x)}{\Delta x} + \epsilon(\Delta x)$$

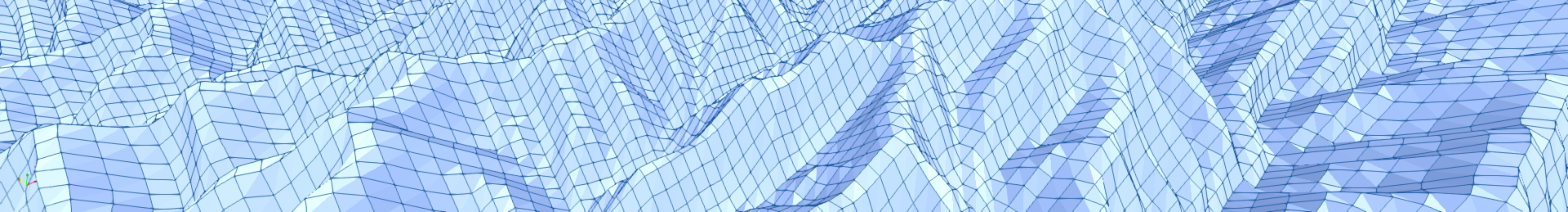
$v > 0$, we must use a forward difference

$$\frac{\partial f}{\partial x} = \frac{f(x) - f(x - \Delta x)}{\Delta x} + \epsilon(\Delta x)$$

$v < 0$, we must use a backward difference

An upwind scheme is the only stable scheme for the SPL (a hyperbolic advection equation)

Centered difference is a mixed forward/backward and, although it is accurate, it is likely to become unstable regardless of the sign of v



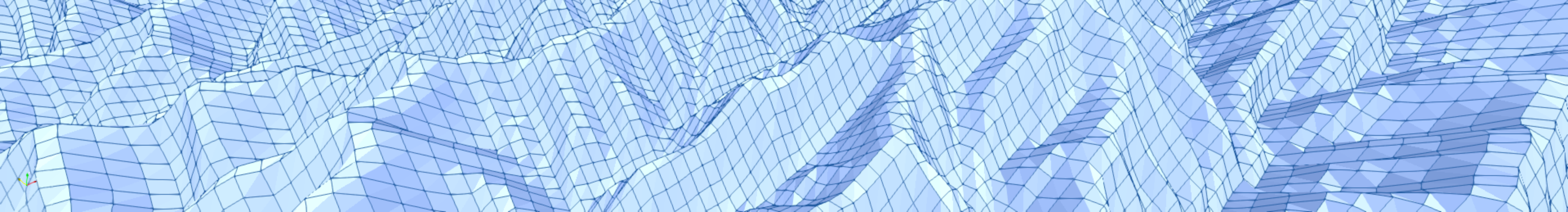
Second derivative

$$(1) \quad f(x + \Delta x) = \sum_{n=0}^{\infty} \frac{\Delta x^n}{n!} \frac{\partial^n f}{\partial x^n} = f(x) + \Delta x \frac{\partial f}{\partial x}(x) + \frac{\Delta x^2}{2} \frac{\partial^2 f}{\partial x^2}(x) + \frac{\Delta x^3}{6} \frac{\partial^3 f}{\partial x^3}(x) + \epsilon(\Delta x^4)$$

$$(2) \quad f(x - \Delta x) = \sum_{n=0}^{\infty} \frac{(-\Delta x)^n}{n!} \frac{\partial^n f}{\partial x^n} = f(x) - \Delta x \frac{\partial f}{\partial x}(x) + \frac{\Delta x^2}{2} \frac{\partial^2 f}{\partial x^2}(x) - \frac{\Delta x^3}{6} \frac{\partial^3 f}{\partial x^3}(x) + \epsilon(\Delta x^4)$$

$$(1+2) \quad \frac{\partial^2 f}{\partial x^2} = \frac{f(x + \Delta x) - 2f(x) + f(x - \Delta x)}{\Delta x^2} + \epsilon(\Delta x^3)$$

Third order accurate approximation of the second derivative



Time integration scheme

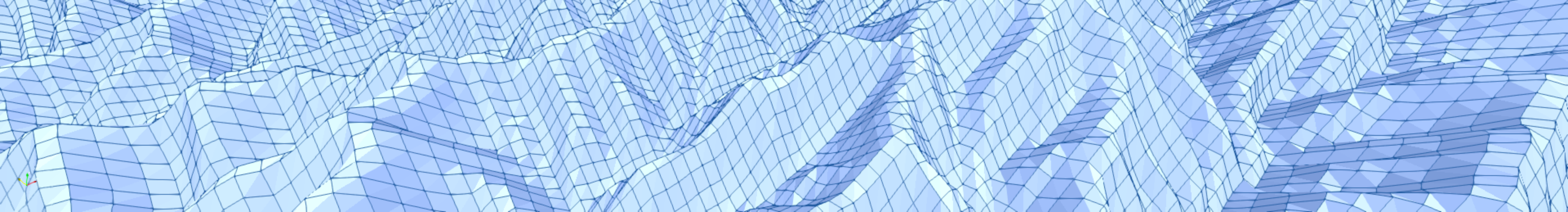
$$\frac{\partial h}{\partial t} = -K_f A^m \frac{\partial h}{\partial x}$$

Explicit scheme

$$\frac{\partial h}{\partial t} = \frac{h(x, t + \Delta t) - h(x, t)}{\Delta t} \approx -K_f A^m \frac{h(x, t) - h(x - \Delta x, t)}{\Delta x}$$

Implicit scheme

$$\frac{\partial h}{\partial t} = \frac{h(x, t + \Delta t) - h(x, t)}{\Delta t} \approx -K_f A^m \frac{h(x, t + \Delta t) - h(x - \Delta x, t + \Delta t)}{\Delta x}$$

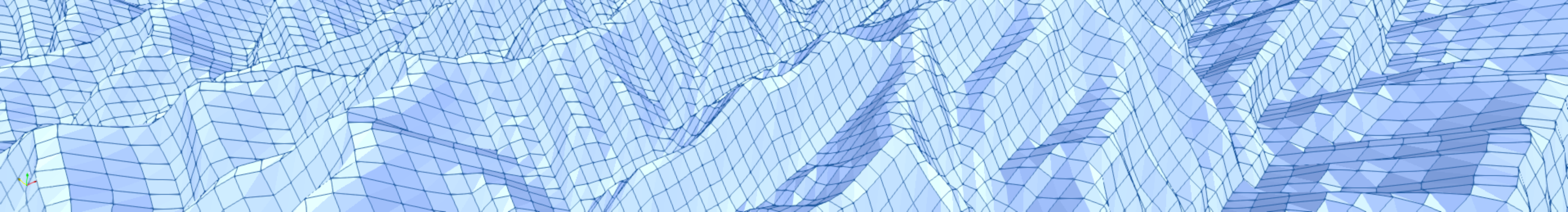


Explicit time integration SPL

$$\frac{h(x, t + \Delta t) - h(x, t)}{\Delta t} \approx -K_f A^m \frac{h(x, t) - h(x - \Delta x, t)}{\Delta x}$$

$$h(x, t + \Delta t) = h(x, t) - \frac{\Delta t K_f A^m}{\Delta x} (h(x, t) - h(x - \Delta x, t))$$

$$h_i = h_i - \frac{\Delta t K_f A_i^m}{\Delta x} (h_i - h_{i-1})$$



Implicit time integration SPL

$$\frac{h(x, t + \Delta t) - h(x, t)}{\Delta t} \approx -K_f A^m \frac{h(x, t + \Delta t) - h(x - \Delta x, t + \Delta t)}{\Delta x}$$

$$h(x, t + \Delta t) = \left(h(x, t) + \frac{\Delta t K_f A^m}{\Delta x} h(x - \Delta x, t + \Delta t) \right) / \left(1 + \frac{\Delta t K_f A^m}{\Delta x} \right)$$

$$h_i = \frac{h_i - F_i h_{i-1}}{1 + F_i} \text{ where } F_i = \frac{\Delta t K_f A_i^m}{\Delta x}$$

Only works if we start from base level !!!